PROBLEM FS2

Hemispherical Station on the Sea Floor

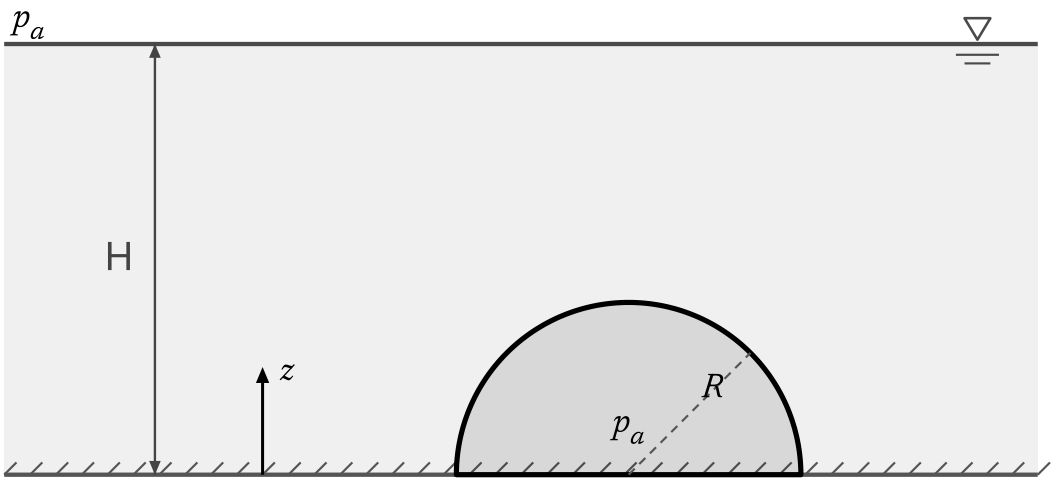
Original statement · fluid_statics_2.pdf

Curved surface | Spherical coordinates | Projection rule | Sheets FS2 and FS8

OFFICIAL PROBLEM STATEMENT

A hemispherical experimental station of radius R is to be installed on the ocean floor at a depth H. Calculate the force, indicating its direction and sense, exerted on the walls of the station, knowing that the pressure inside is equal to the atmospheric pressure, p_a.

Translated from the Spanish original.



Statement figure (redrawn from fluid_statics_2.pdf and fluid_statics_8.pdf, p. 1): hemispherical station of radius R on the sea floor at depth H, atmospheric pressure p_a inside, axis z up from the floor

Source: fluid_statics_2.pdf and fluid_statics_8.pdf (page 1, sheet of 29-01-01). Notes.pdf, Chapter 5: Eqs. 5.10, 5.11 and 5.14.

Worked solution

Hypotheses, Data and Unknowns

- Sea water at rest, incompressible, density ρ; the free surface is at the atmospheric pressure p_a, the depth of the floor is $H > R$.
- The walls form a thin hemispherical shell of radius R resting on the floor; inside, the air is at p_a. The force asked for is the net pressure force on the shell (water outside plus air inside); the floor of the station is not part of the walls.
- Data: R, H, ρ, g, p_a. Unknown: the force $\vec{F}$.

Symbol	Meaning	SI unit
R	radius of the station	m
H	depth of the sea floor	m
ρ	density of sea water	kg/m^3

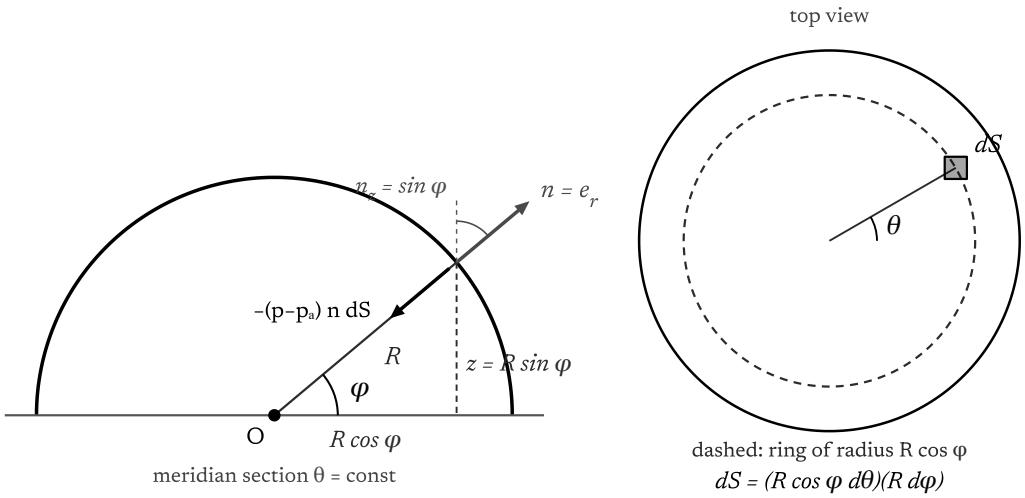
Coordinates, Surface Element and Normal

Take the origin O at the centre of the base of the hemisphere, z vertical and pointing up, so the floor is $z = 0$ and the free surface $z = H$. A point of the shell is located by its **elevation** φ above the floor plane ($0 \le \varphi \le \pi/2$) and its **azimuth** θ ($0 \le \theta < 2\pi$). In the meridian section (right triangle with the radius R as hypotenuse), the height $z = R\sin\varphi$ is the leg opposite to φ and the distance to the axis $R\cos\varphi$ the adjacent leg.

The element of the sphere is a small rectangle with sides $R\,d\varphi$ (along the meridian) and $R\cos\varphi\,d\theta$ (along the parallel, a circle of radius $R\cos\varphi$):

$$dS = R^2\cos\varphi\,d\theta\,d\varphi$$

The unit normal towards the water is the radial vector $\vec{e}_r = \cos\varphi\cos\theta\,\vec{e}_x + \cos\varphi\sin\theta\,\vec{e}_y + \sin\varphi\,\vec{e}_z$; its vertical component $n_z = \sin\varphi$ is the leg opposite to φ of the unit right triangle.



Spherical coordinates of the station. Left: meridian section; the radius R is the hypotenuse, $z = R\sin\varphi$ is the leg opposite to the elevation φ and $R\cos\varphi$ the adjacent leg; the outward normal is $\vec{e}_r$, with $n_z = \sin\varphi$; the water pushes along $-\vec{e}_r$. Right: top view; at elevation φ the parallel is a ring of radius $R\cos\varphi$, so the element is $dS = (R\cos\varphi\, d\theta)(R\, d\varphi)$

The Pressure and the Net Elementary Force

By (5.10), at the height z (depth $H - z$): $p = p_a + \rho g(H - z) = p_a + \rho g(H - R\sin\varphi)$.

The water pushes the outer face with $-p\,\vec{e}_r\,dS$; the air pushes the inner face with $-p_a(-\vec{e}_r)\,dS$, since the normal towards the air is $-\vec{e}_r$. Their sum is

$$d\vec{F} = -(p - p_a)\,\vec{e}_r\,dS = -\rho g(H - R\sin\varphi)\,\vec{e}_r\,R^2\cos\varphi\,d\theta\,d\varphi$$

Integration

Horizontal components. F_x contains $\int_0^{2\pi}\cos\theta\,d\theta = [\sin\theta]_0^{2\pi} = 0$, and F_y contains $\int_0^{2\pi}\sin\theta\,d\theta = [-\cos\theta]_0^{2\pi} = 0$: by the axial symmetry the horizontal forces cancel.

Vertical component. With $n_z = \sin\varphi$, the θ integral of a function independent of θ gives 2π:

$$F_z = -\int_0^{\pi/2}\int_0^{2\pi} \rho g(H - R\sin\varphi)\sin\varphi\, R^2\cos\varphi\, d\theta\, d\varphi = -2\pi R^2\rho g \int_0^{\pi/2} \left(H\sin\varphi\cos\varphi - R\sin^2\varphi\cos\varphi\right) d\varphi$$

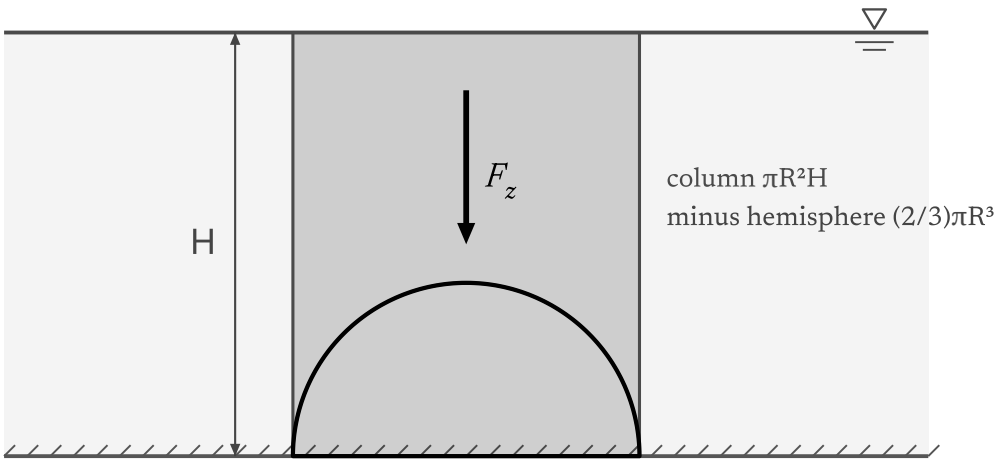
Both terms are integrated with the substitution $u = \sin\varphi, du = \cos\varphi\, d\varphi$, with $u = 0$ at $\varphi = 0$ and $u = 1$ at $\varphi = \pi/2$:

$$\int_0^{\pi/2} \sin\varphi\cos\varphi\, d\varphi = \int_0^1 u\, du = \left[\frac{u^2}{2}\right]_0^1 = \frac{1}{2}, \qquad \int_0^{\pi/2} \sin^2\varphi\cos\varphi\, d\varphi = \int_0^1 u^2\, du = \left[\frac{u^3}{3}\right]_0^1 = \frac{1}{3}$$

Therefore

$$F_z = -2\pi R^2\rho g\left(\frac{H}{2} - \frac{R}{3}\right) = -\pi R^2\rho g\left(H - \frac{2R}{3}\right)$$

Check with the Projection Rule



Projection rule (Eq. 5.14): F_z is the weight of the water that would stand above the hemisphere, the cylinder $\pi R^2 H$ minus the hemisphere $\frac{2}{3}\pi R^3$; it points down

By (5.14), the vertical force on a surface with the water above it is the weight of the water column standing on it, down to the surface: the cylinder of radius R and height H, $\pi R^2 H$, minus the volume of the hemisphere, $\frac{2}{3}\pi R^3$, which is not water:

$$F_z = -\rho g \left(\pi R^2 H - \frac{2}{3}\pi R^3 \right) = -\pi R^2 \rho g \left(H - \frac{2R}{3} \right)$$

the same result. The air inside contributes nothing net because p_a also acts at the top of the column.

Answers and Checks

- $\vec{F} = -\pi R^2 \rho g \, (H - 2R/3) \, \vec{e}_z$: vertical, downward (since $H > R > 2R/3$), pressing the station against the floor; no horizontal component.
- **Limiting case** of a small station, $R \ll H$: $F_z \approx -\rho g H \, \pi R^2$, the gauge pressure at the floor times the base area, as for a flat lid; the term $2R/3$ comes from the hemisphere volume, which is not water.
- **Physical reading**: the station is not lifted by Archimedes' principle, because no water reaches its base: the buoyancy formula (5.15) applies only to a surface closed by water.
- **Dimensions**: $\rho g R^2 H$ is (kg m^{-3})(m s^{-2})(m^3) = N.
- **Check against the official key**: fluid_statics_2.pdf (and its second scan fluid_statics_8.pdf) sets up the same integral, $F_z = -\iint \rho g (H - R\sin\varphi) \sin\varphi \, R^2 \cos\varphi \, d\theta \, d\varphi$, and reaches $-2\pi R^2 \rho g [H \sin^2\varphi/2 - R\sin^3\varphi/3]_0^{\pi/2}$, which equals the result above.

Discrepancy with the official solution

The handwritten last line reads $-\pi R^2 \rho g \, [H - 2R/3] \, \vec{e}_z$ preceded by a mark that looks like a struck-out 4 (reading of handwriting). The previous line of the key, $-2\pi R^2 \rho g \, [H \sin^2\varphi/2 - R\sin^3\varphi/3]_0^{\pi/2}$, evaluates to $-2\pi R^2 \rho g \, (H/2 - R/3) = -\pi R^2 \rho g \, (H - 2R/3)$, coefficient π, as above.